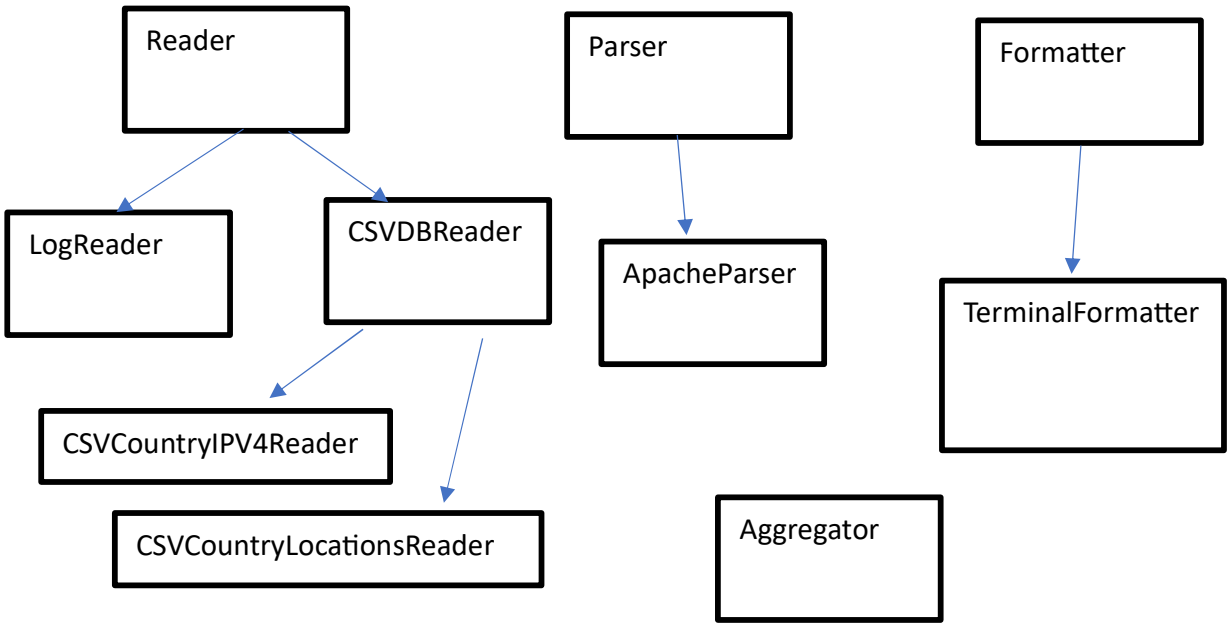
Technical Design- Yarin Gdanski

High-level architecture and component breakdown:



Key abstractions and interfaces:

- An abstract class for reading a file, and a class for reading from a log file and a db file
- An abstract class for printing an output and a class for outputting in the specified format in the terminal
- An abstract parser and a class for parsing an apache file

How you will handle extensibility (adding new dimensions, output formats, etc.):

In order to handle new dimensions, I am using a constant dictionary, in which the fields I would like to extract from a line will be defined. Each item will include the field name and the index of it in the line in the apache log.

In order to support different output formats I am using an abstract method to output data in a generic format, and if one will want to add a capability to output to a file or in a specific order, he would do so.

Technology choices and rationale:

I am using the Python language and the module- ua_parser to parse the UserAgent from each line. In order to read a csv DB I am using the csv module.

Any trade-offs or assumptions:

I assume the Apache file I am receiving is not corrupted, and that the IP address is in a correct format, like all other fields.

Trade-offs- We are limited in the read speed: more lines and more dimesions, will add more fields in the dictionary and more time to read everything.